

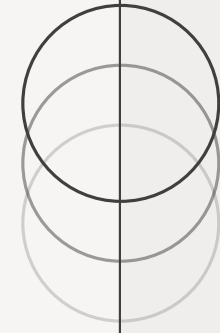
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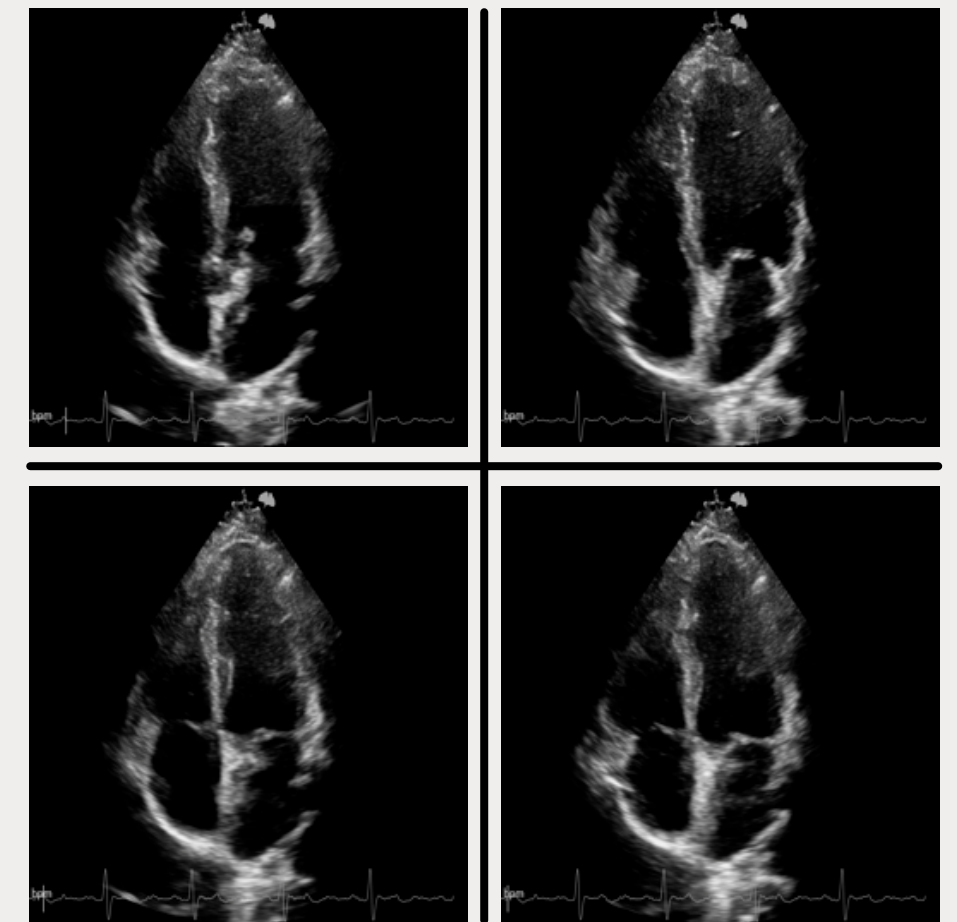
AI-DRIVEN DEHAZING AND DENOISING OF ECHOCARDIOGRAPHIC IMAGES USING MAMBA

MOTIVATION



Why this problem matters

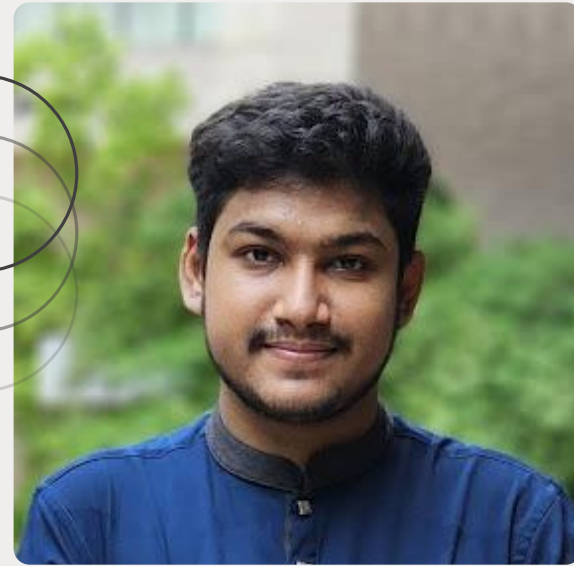
- Echocardiography is operator dependent
- Image quality degrades due to:
 - Patient anatomy (obesity, lung interference)
 - Acoustic shadowing
 - Speckle noise
- Old Machinery



Poor image quality → missed or delayed diagnosis

THE TEAM

03



Name : Nusrat Jahan Niha

ID : 2221787042

Contribution : focused on data preparation, including preprocessing, augmentation, and exploratory analysis, and implemented U-Net as the baseline model for denoising and dehazing.



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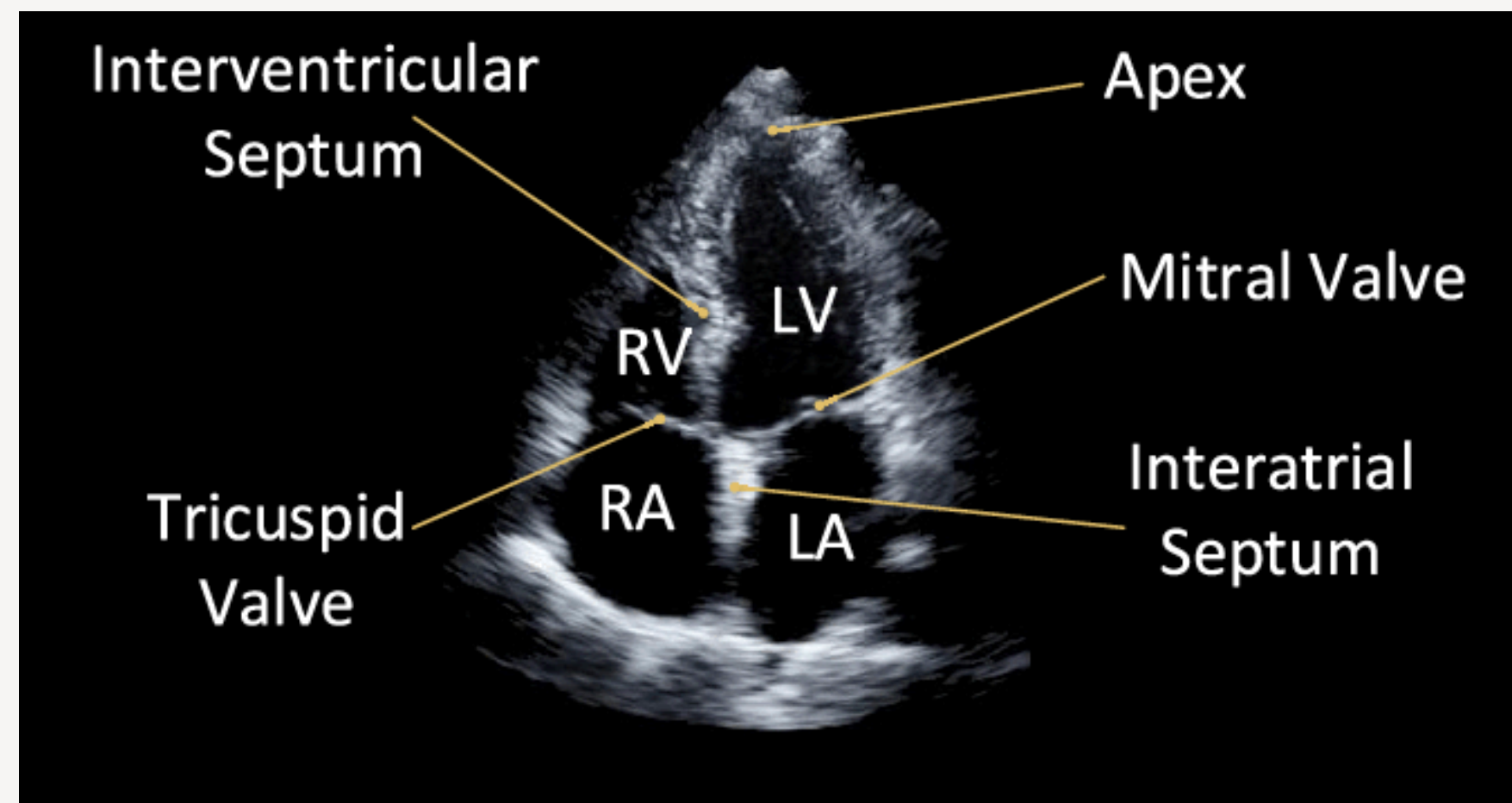
Name : Aritra Islam Saswato

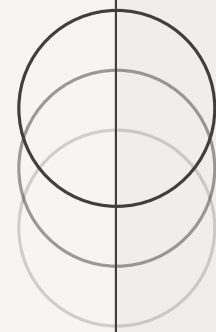
ID : 2132629642

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Clinical Problem Statement

- Cardiologists rely on visual clarity
- Low-quality echo images affect:
 - Chamber boundary visibility
 - Valve morphology
 - Wall motion assessment
- Existing hardware-based solutions are:
 - Expensive
 - Non-adaptive
 - Not patient-specific

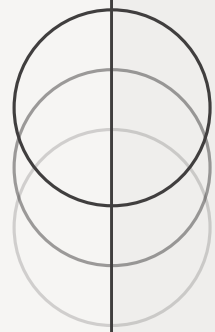




RESEARCH GAP

What is missing in current Research

- Most prior works :
 - Focus only on pixelwise denoising OR dehazing
 - Ignore clinical metrics
- Limited comparison between :
 - GANs
 - Diffusion models
- Lack of real clinical dataset validation
- No implementation using Mamba

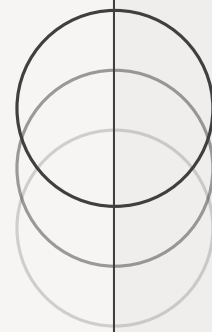


Research Objectives

- Primary Objective :
 - Enhance echocardiographic image quality using deep learning
- Secondary Objectives :
 - Compare GAN, Diffusion, and Mamba models
 - Preserve clinical structures
 - Improve diagnostic confidence



DATASET OVERVIEW



Data Source

Dehazing Echocardiography
Challenge 2025¹

Dataset Summary

- 75 patients
- 4,376 high-quality reference images
- 2,324 degraded images
- 237 ROI annotations

1. <https://dehazingecho2025.grand-challenge.org/>

Data Challenges

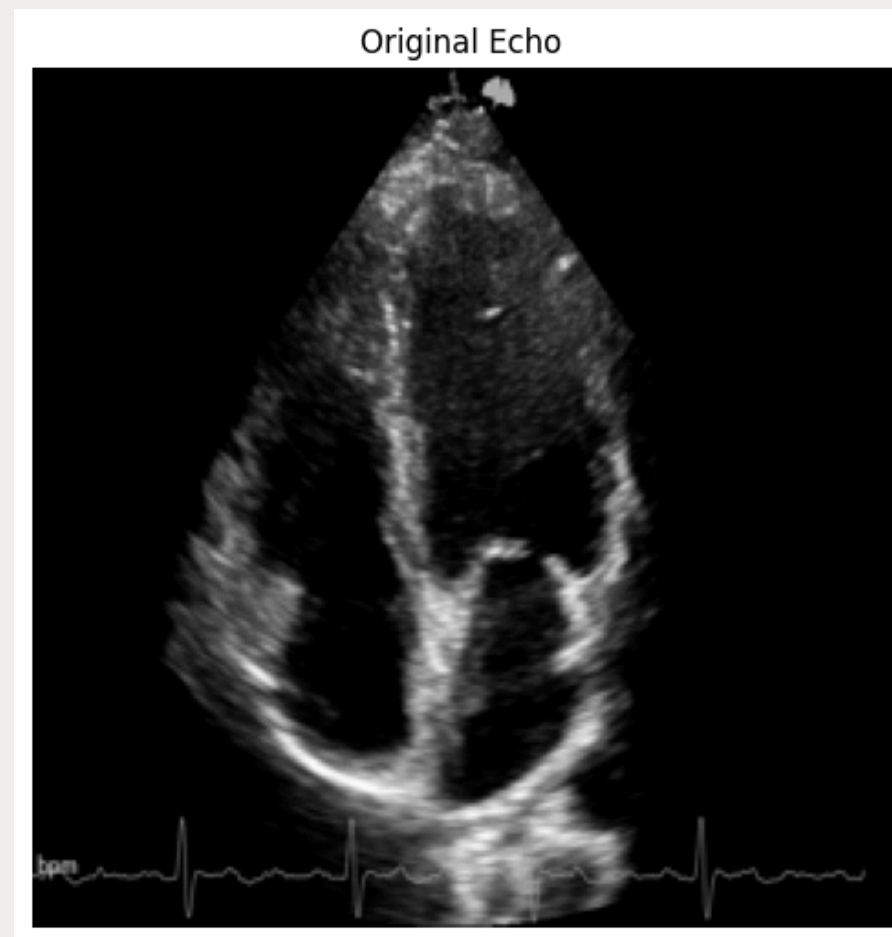
- Lack of noisy-clean data pairs
- Non-uniform noise
- View dependency
- Patient variability

Data Preprocessing Pipeline

- Image resizing & normalization
- Intensity standardization
- Data augmentation :
 - Flipping
 - Rotation
 - Contrast variation

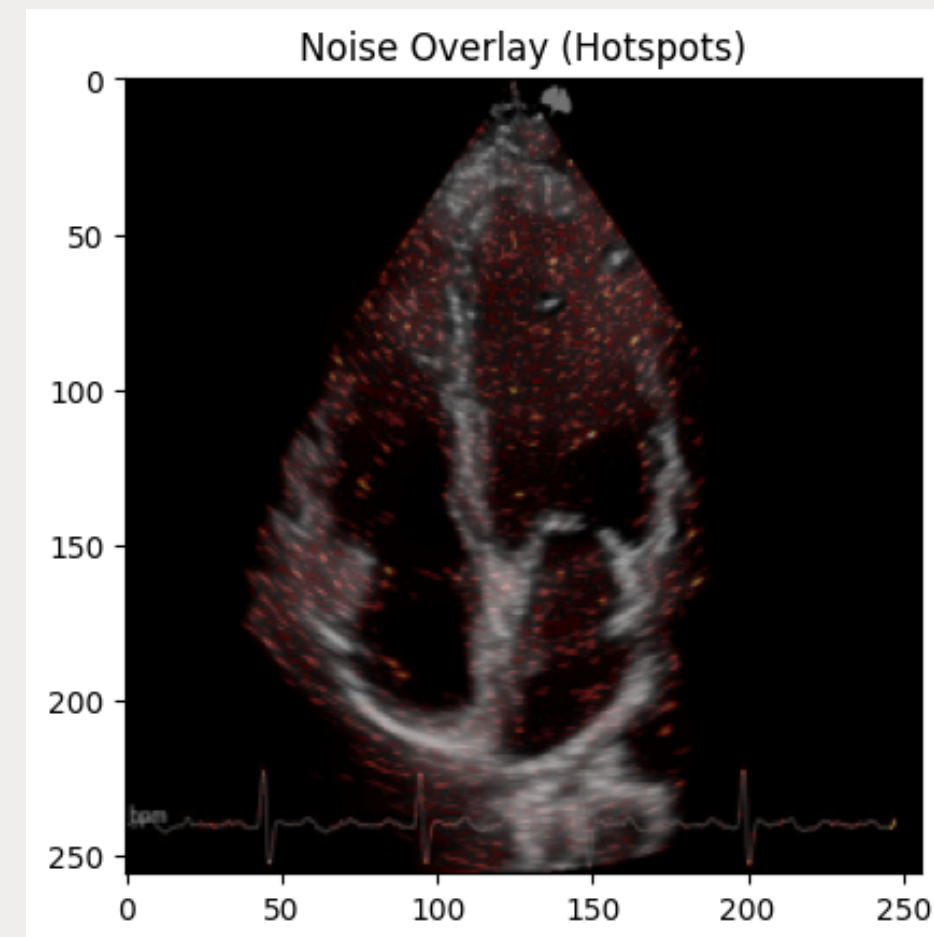


Problem with Noisy Data



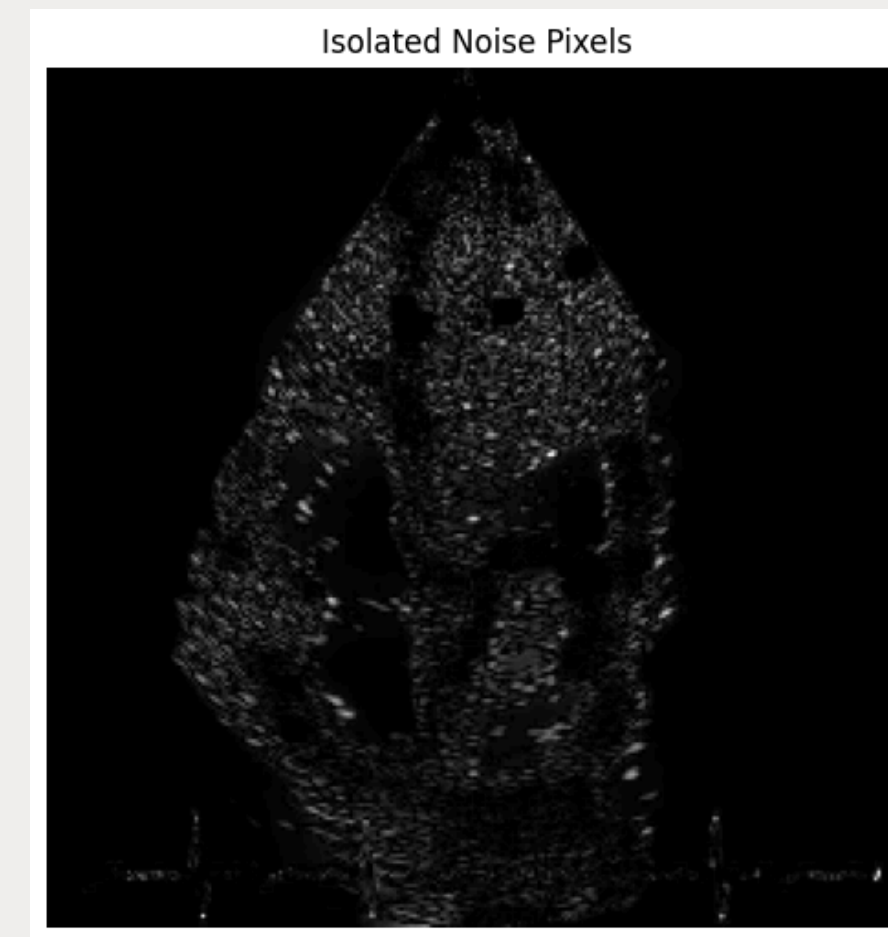
Original Noisy Data

- Real clinical echocardiogram
- Contains speckle noise and haze
- Low contrast in critical regions



Noise Overlay (Hotspots)

- High-intensity noise regions showed
- Shows spatial concentration of noise
- Noise overlaps with anatomical structures



Isolated Noise Pixels

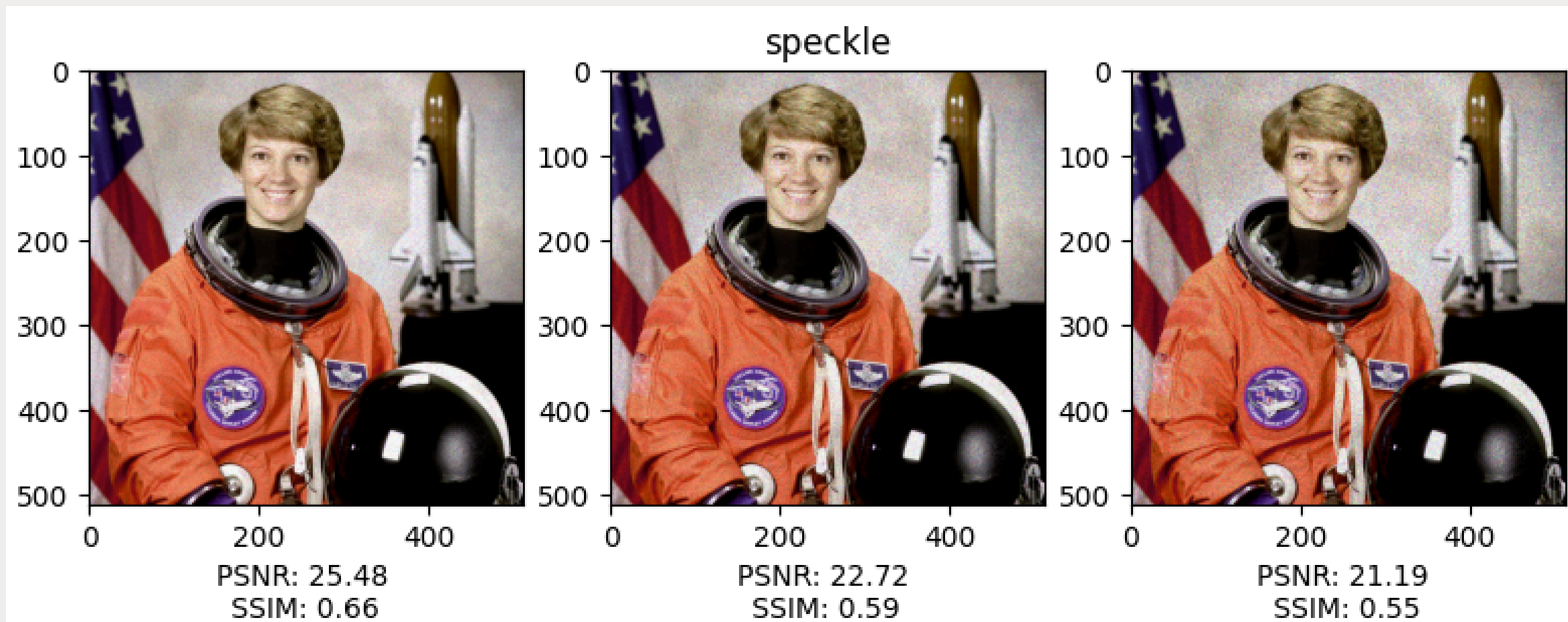
- Extracted noise pixels from anatomical structures
- Confirms noise is not random
- Better visualize what needs to be removed

Evaluation Metrics

PSNR (Peak Signal-to-Noise Ratio) : Fast but often fails to align with human perception

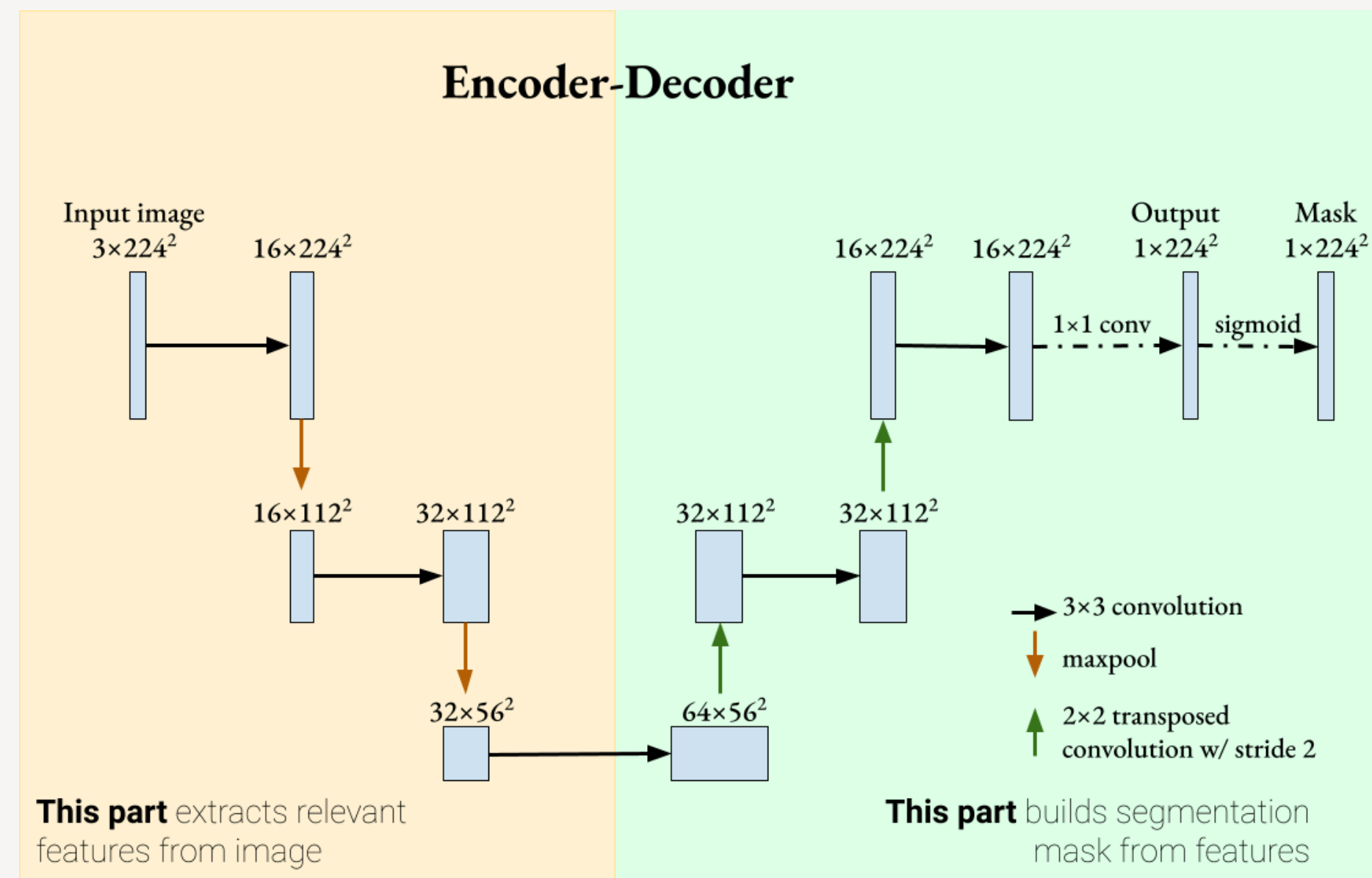
SSIM (Structural Similarity Index Measure): It is better for perceived quality

CNR (Contrast-to-Noise Ratio): critical for diagnostic imaging.



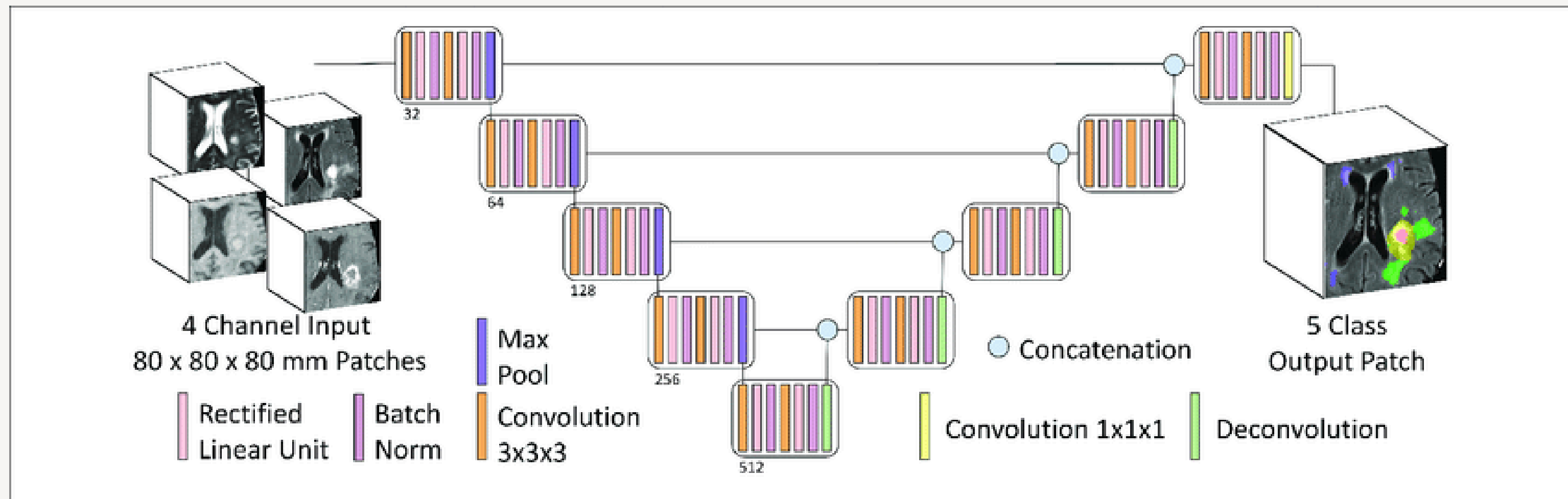
U-Net: How It Works

- Encoder-decoder architecture with symmetric skip connections
- Encoder extracts hierarchical features from noisy input
- Decoder reconstructs enhanced image from compressed representation
- Skip connections preserve spatial detail lost during downsampling
- Widely used as a baseline in medical image enhancement



U-Net in Our Framework

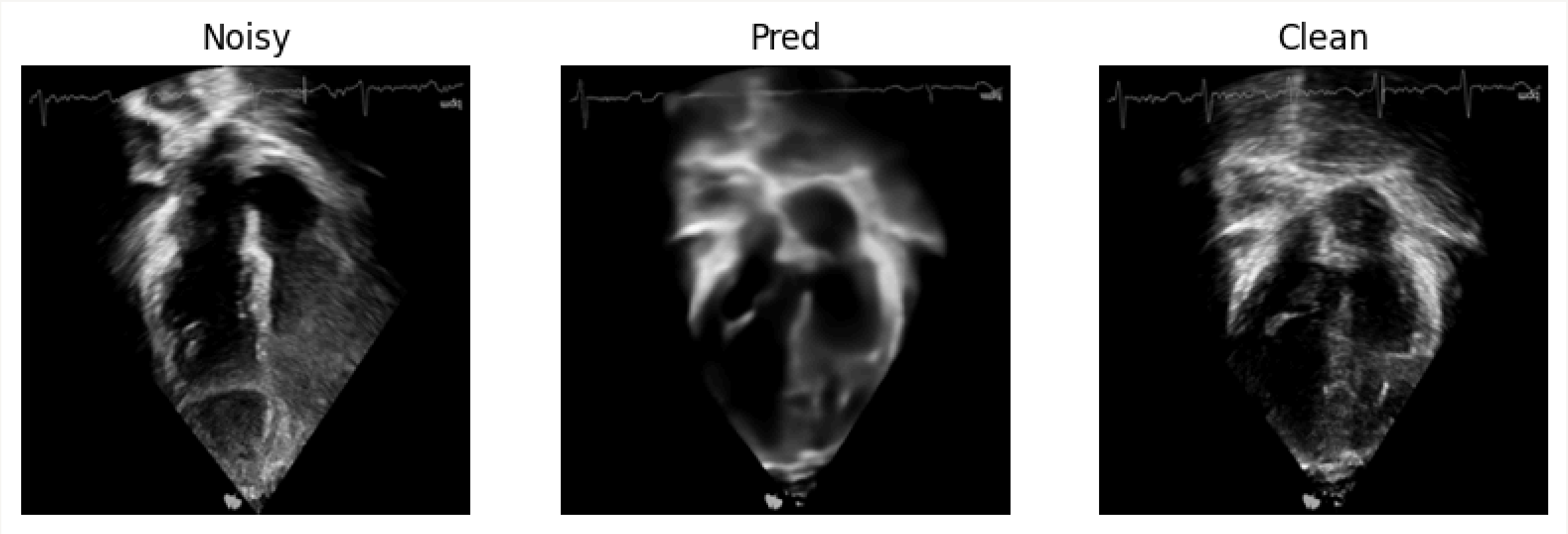
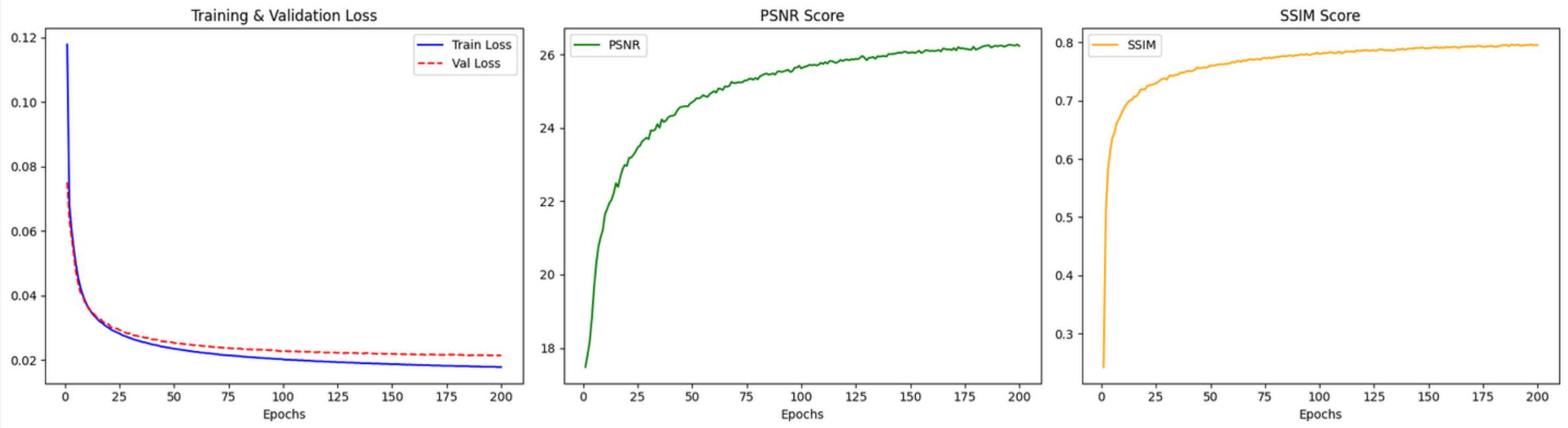
- Used as a baseline denoising and dehazing model
- Processes echocardiograms in an end-to-end manner
- Optimized using pixel-level reconstruction loss
- Also serves as the generator backbone in GAN experiments
- Provides a reference point for more advanced models



U-Net Results

FINAL TRAINING REPORT

Metric	Epoch	Value
Best PSNR	199.0	26.270
Best SSIM	199.0	0.796



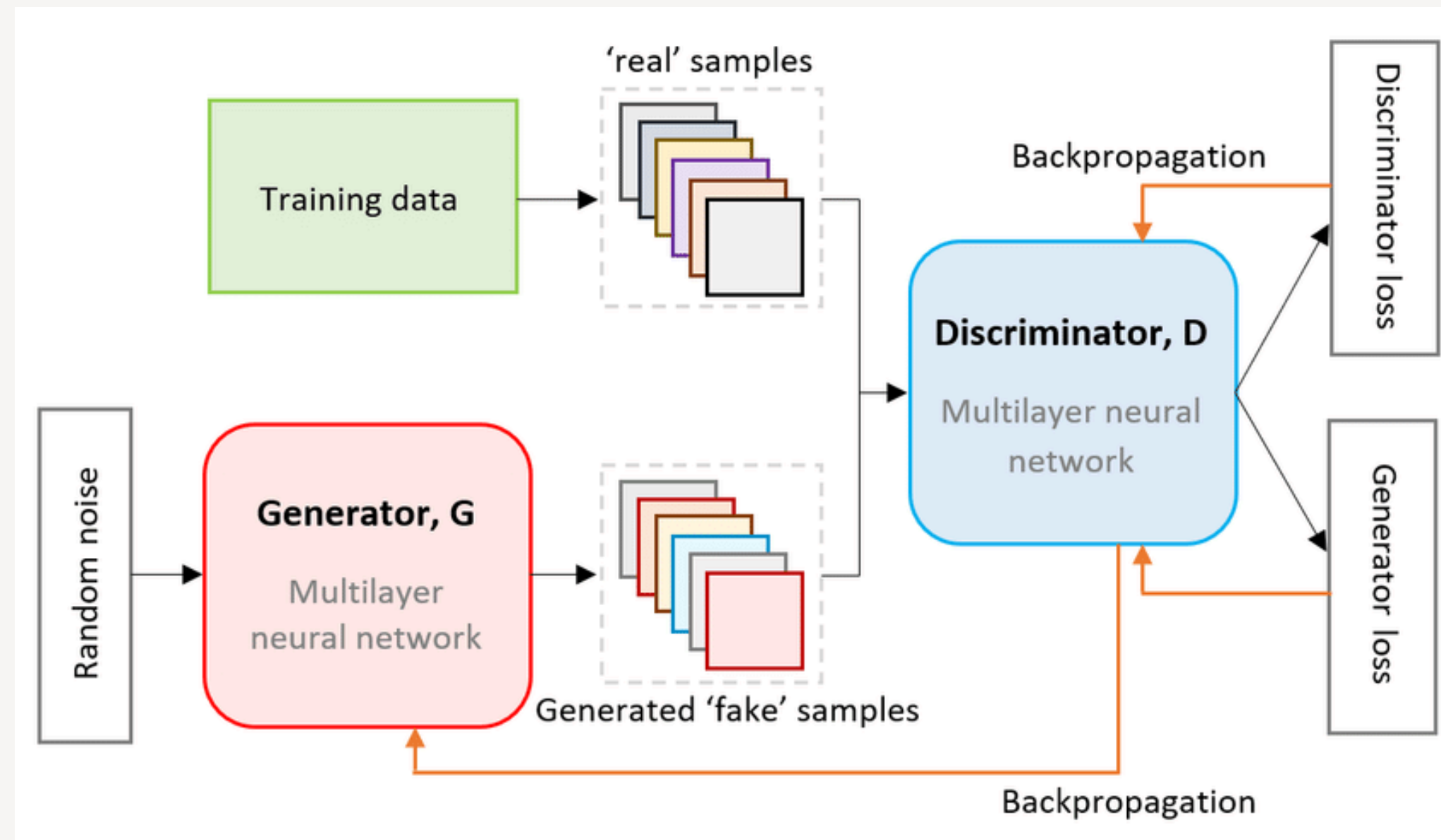
Limitations of Classical CNN-Based Enhancement

- Classical CNNs (e.g., U-Net) rely on local receptive fields, limiting their ability to capture global spatial context in echocardiographic images.
- Echocardiographic noise is structured and overlaps with anatomy, making simple convolution-based filtering unreliable.
- CNN-based enhancement often leads to over-smoothing, where clinically important boundaries and fine structures are lost.
- Pixel-level optimization alone is insufficient to ensure perceptual realism and clinical safety.
- Therefore, a single model family cannot address all aspects of the problem.

To balance noise suppression, structural preservation, and clinical relevance, we explore multiple deep learning paradigms, progressing from CNNs to GANs, diffusion models, and state-space architectures.

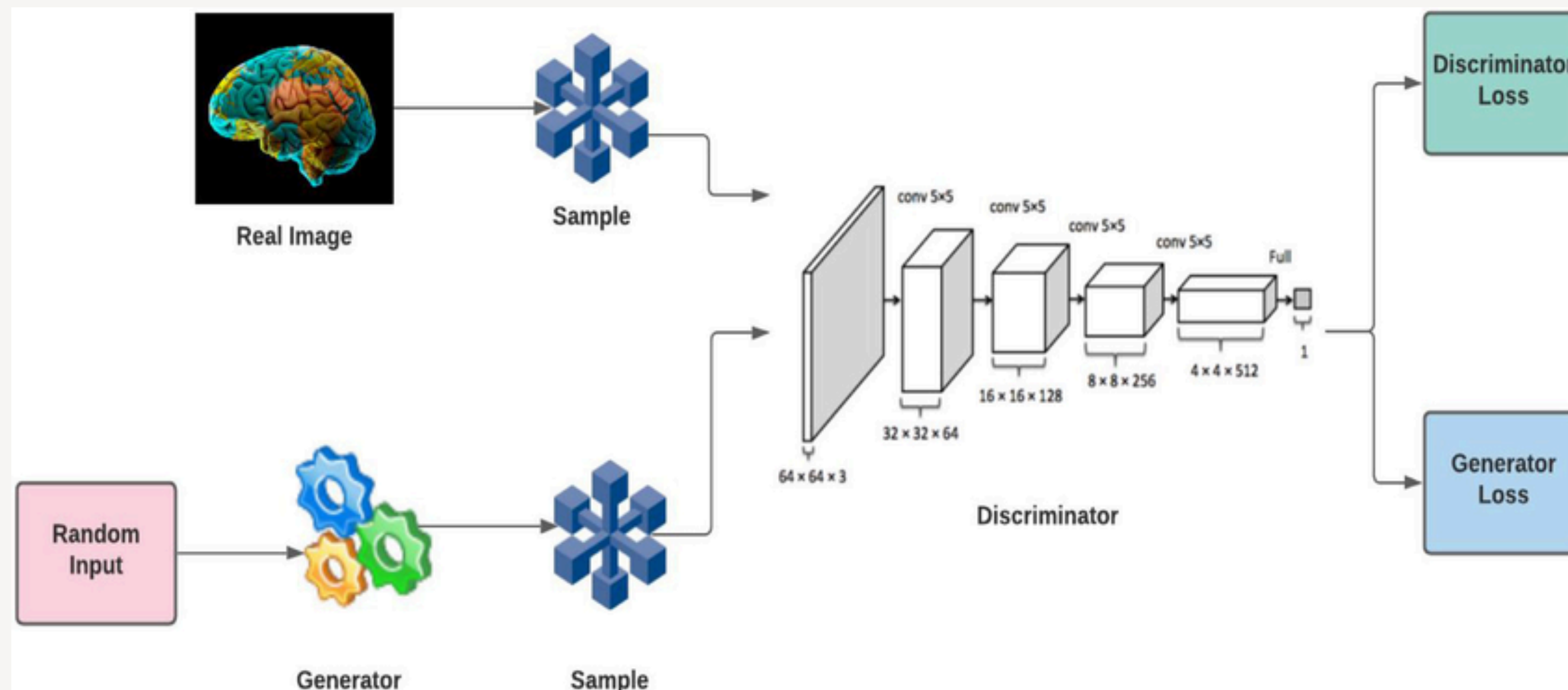
GAN: How It Works

- Composed of a generator and a discriminator
- Generator aims to produce realistic enhanced images
- Discriminator distinguishes real from generated images
- Adversarial training encourages perceptual realism
- Learns complex image distributions beyond pixel similarity

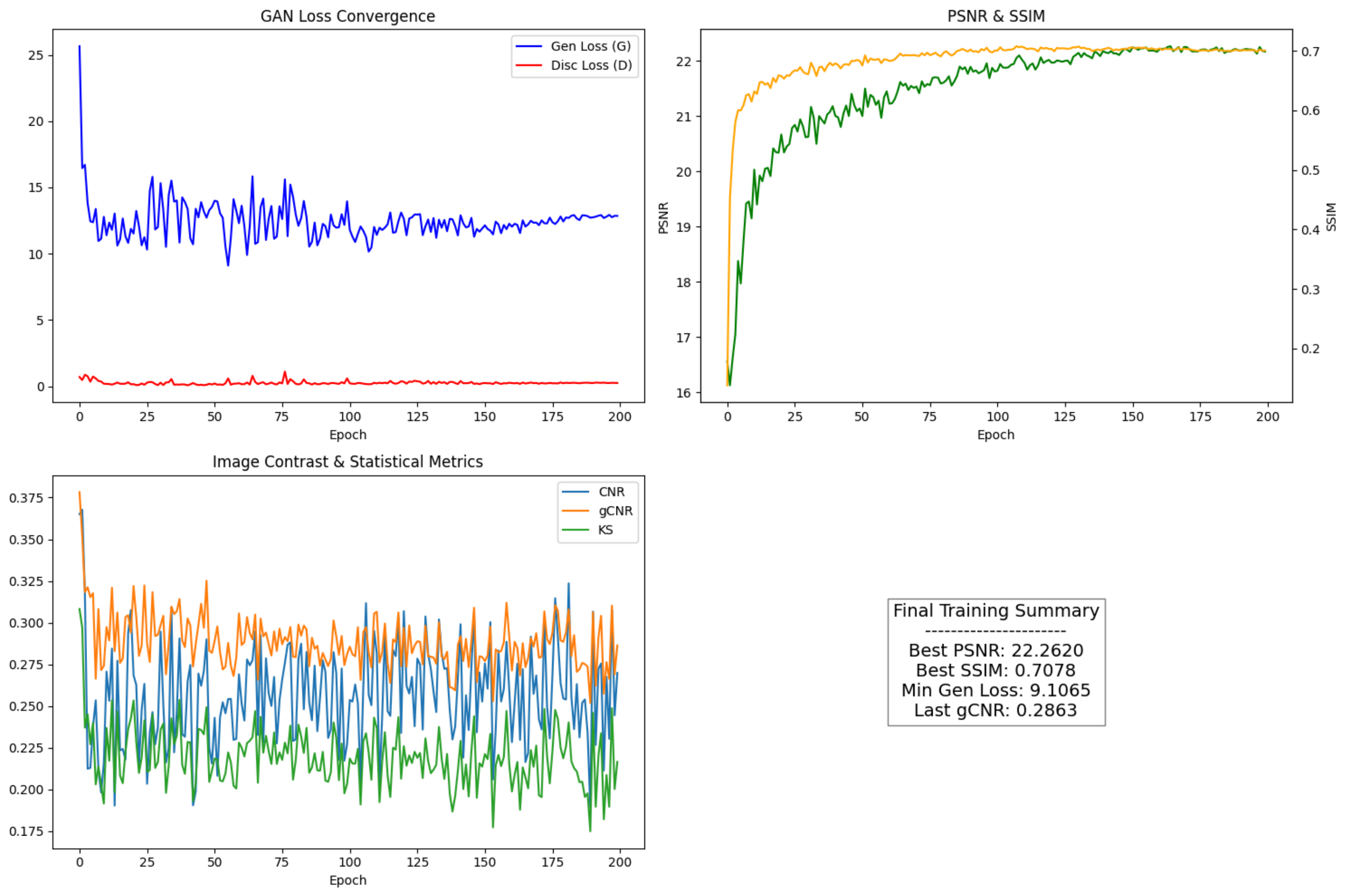


GAN Architecture in Our Study

- U-Net used as the generator network
- PatchGAN used as the discriminator
- Combined loss: adversarial + L1 loss
- Patch-based discrimination focuses on local texture realism
- Designed to preserve fine anatomical details



GAN Results



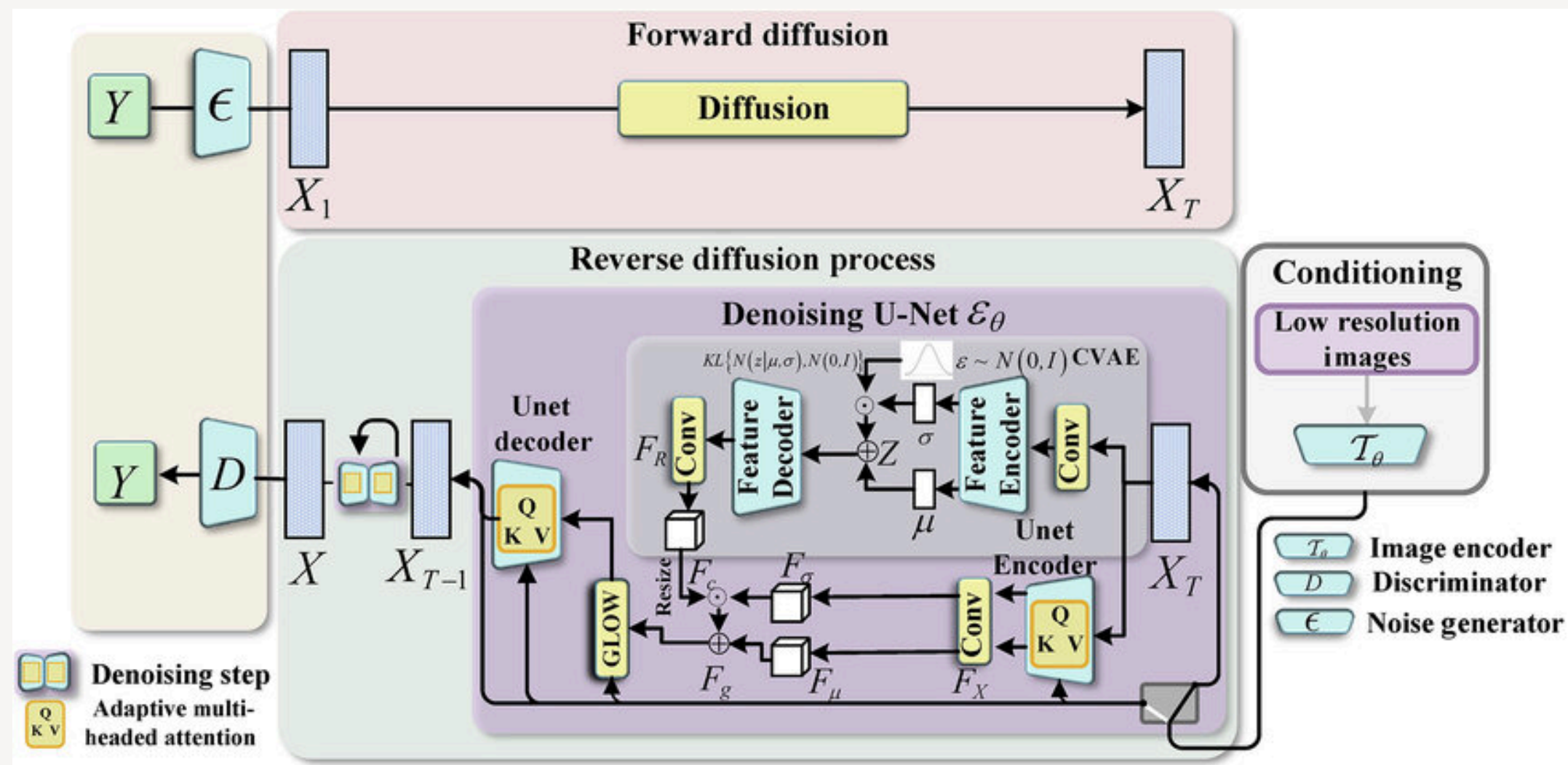
Limitations of GAN

- Training instability and sensitivity to hyperparameters
- Risk of hallucinating structures not present in input
- Hard to guarantee clinical safety
- Performance varies across patients
- Less suitable for deployment without strong constraints

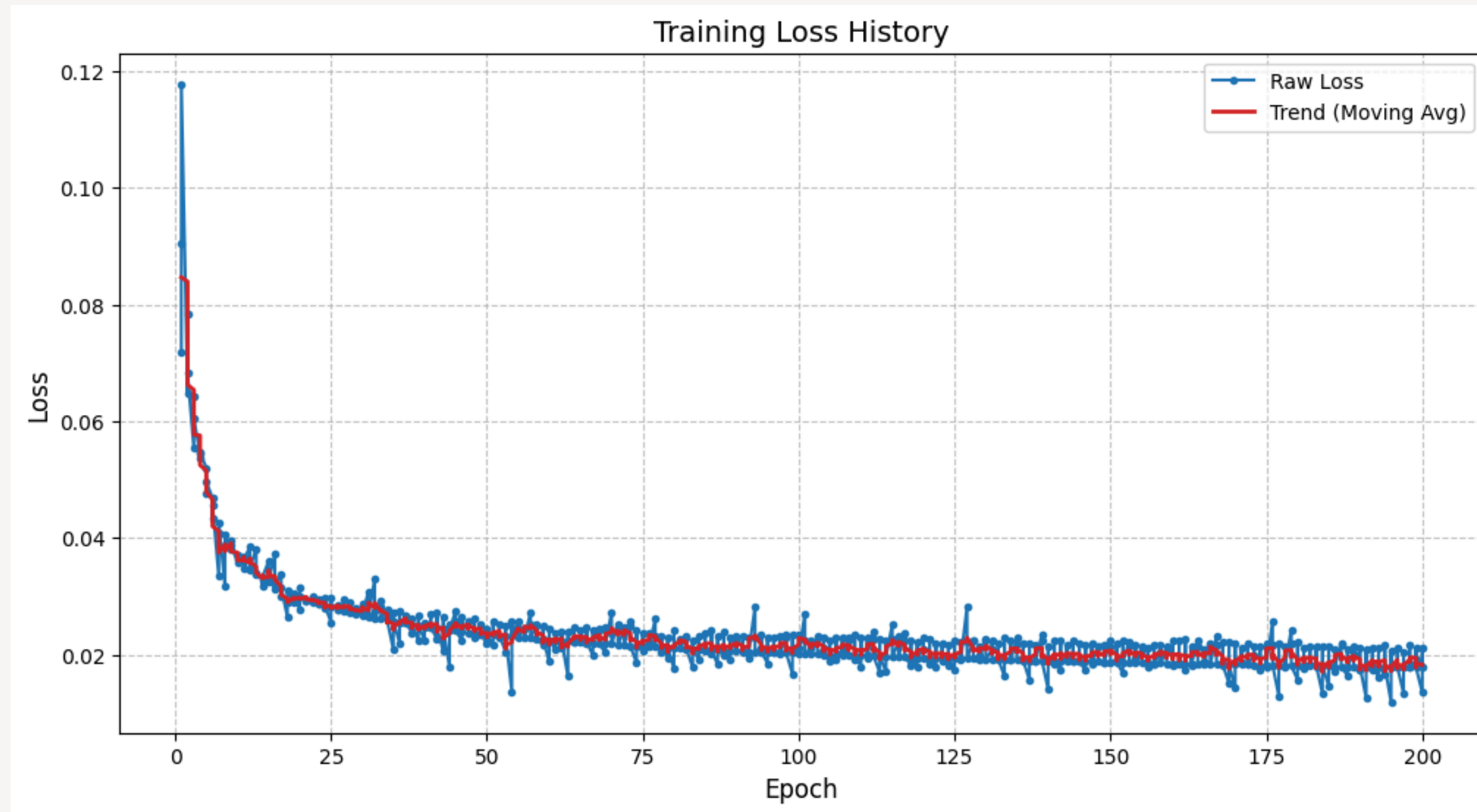
GANs improve realism but compromise reliability; therefore, we explore models designed for stability and structure preservation.

Diffusion Models: Working Principle

- Forward process gradually adds noise to images
- Model learns to reverse the noise process step by step
- Generates images through probabilistic denoising
- Training is stable and well-behaved
- Strong theoretical guarantees for structure preservation



Diffusion Training History



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TRAINING FINAL RESULTS

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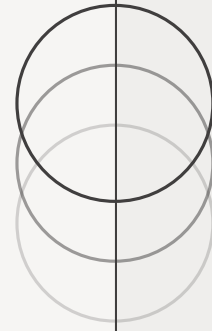
Total Epochs: 200
Best (Min) Loss: 0.011900
Final Loss: 0.017800

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Our full training could not be completed due to resource limitation

Strength & Limitations



Strength

Produces structurally consistent enhancements, less prone to hallucination, preserves global anatomical structures.

Limitations

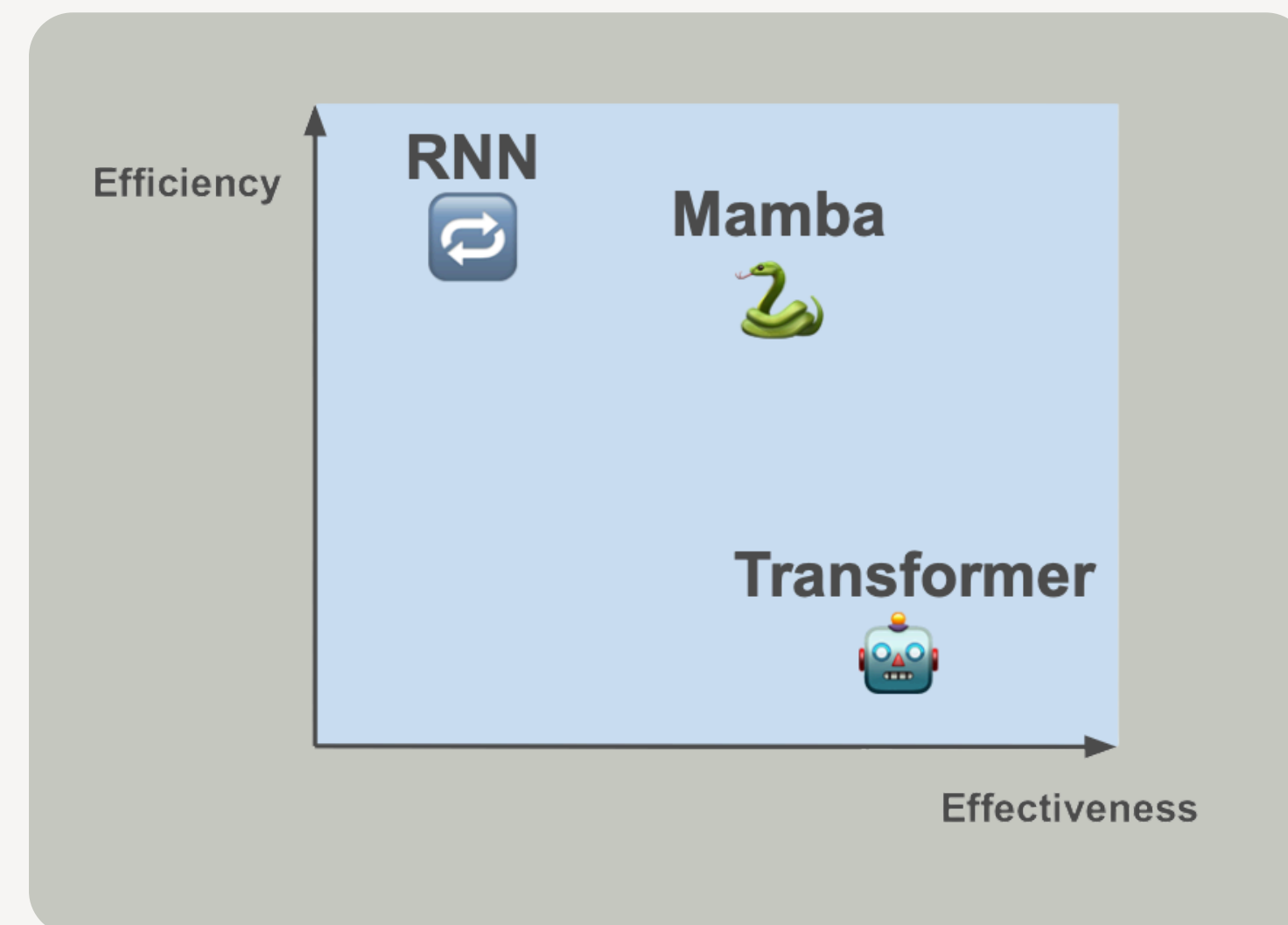
Outputs are less sharp than GANs, and inference is slow due to computationally expensive iterative steps, limiting real-time use.

Motivation for Mamba

- Need for global context awareness in echocardiographic images
- Avoid hallucination while preserving structure
- Require stable and efficient training
- CNNs are local, GANs unstable, Diffusion slow and resource consuming
- State-space models offer a promising alternative

What Problem Mamba Solves

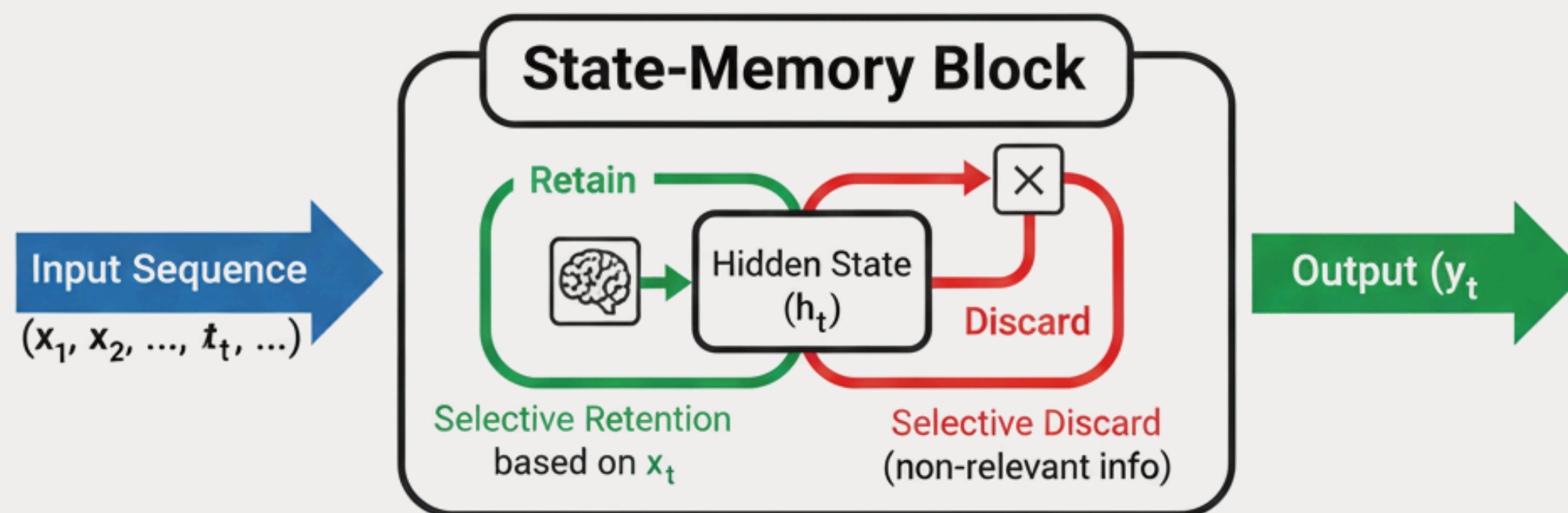
- Captures long-range spatial dependencies efficiently
- Avoids quadratic complexity of attention mechanisms
- More stable than adversarial training
- Balances local and global feature learning
- Suitable for high-resolution medical images



Intuition Behind Mamba



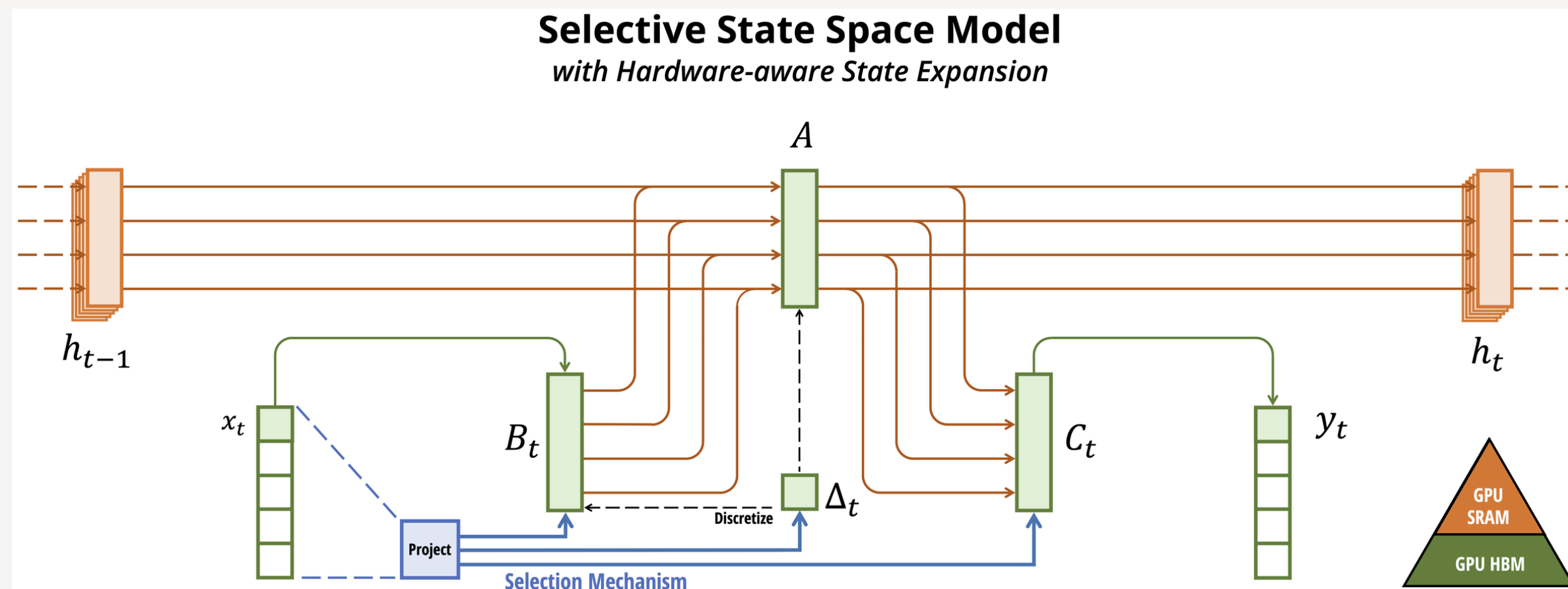
- Treats image features as sequential signals
- Maintains an internal state to capture dependencies
- Selectively retains important information
- Suppresses irrelevant or noisy components
- Functions like a memory-based filtering mechanism



At each step ' t ', the block selectively updates its hidden state, deciding what information from x_t to **RETAIN** and what to **DISCARD**.

Mamba Model Architecture

- Input features passed through state-space layers
- Selection mechanism updates internal states
- Linear computational complexity w.r.t. input size
- No attention mechanism required
- Efficient and scalable design



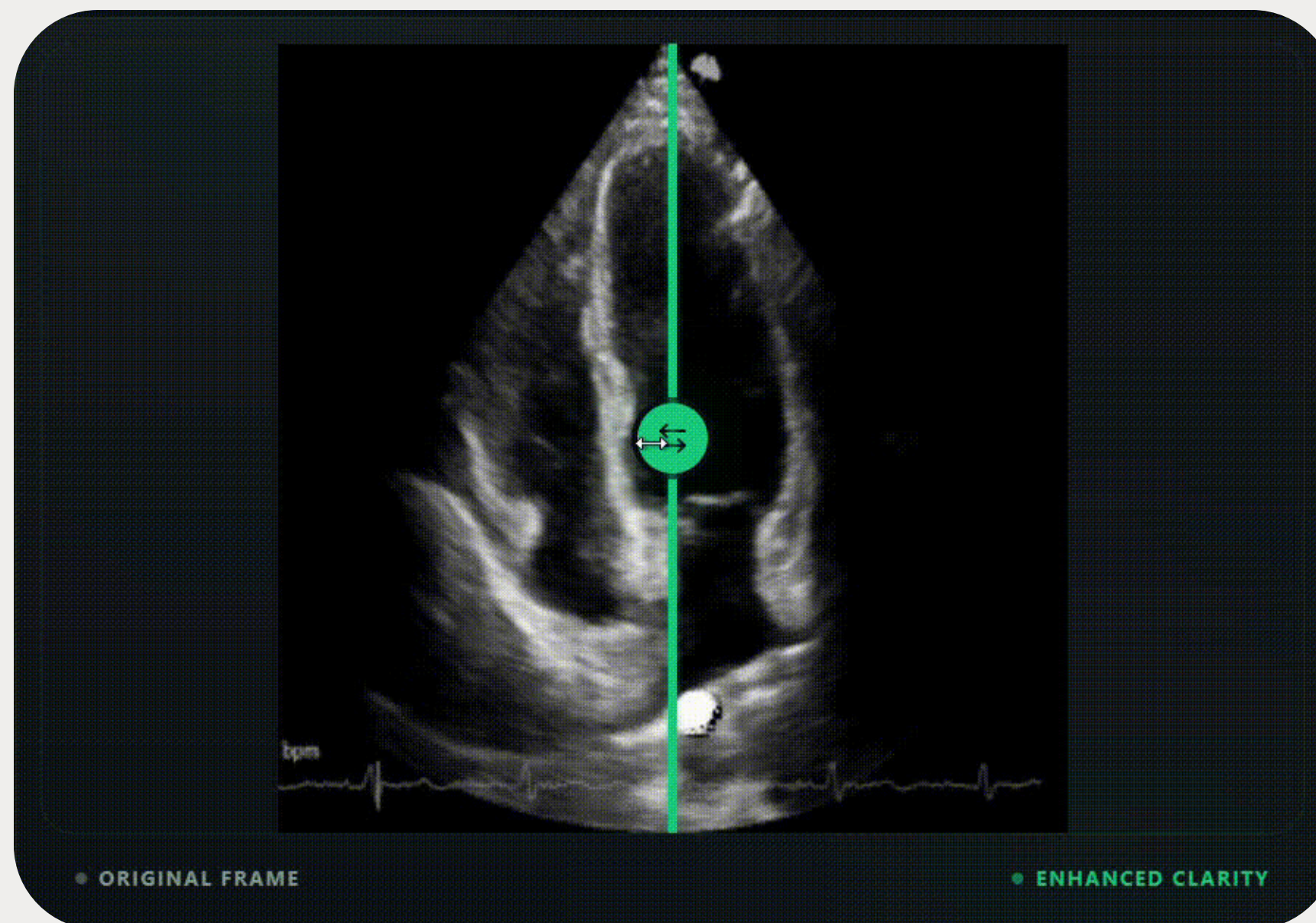
Mamba: Linear-Time Sequence Modeling with Selective State Spaces

Authors: Albert Gu*, Tri Dao*

Paper:
<https://arxiv.org/abs/2312.00752>

Why Mamba Fits Echocardiography

- Noise is spatially structured and globally dependent
- Cardiac structures require long-range consistency
- Mamba preserves anatomical continuity
- Reduced risk of artificial texture generation



Our Mamba-Based Framework

- Echocardiograms divided into patch embeddings
- Features processed through stacked Mamba blocks
- Reconstruction head generates enhanced image
- End-to-end training on paired data
- Designed for structure-preserving enhancement

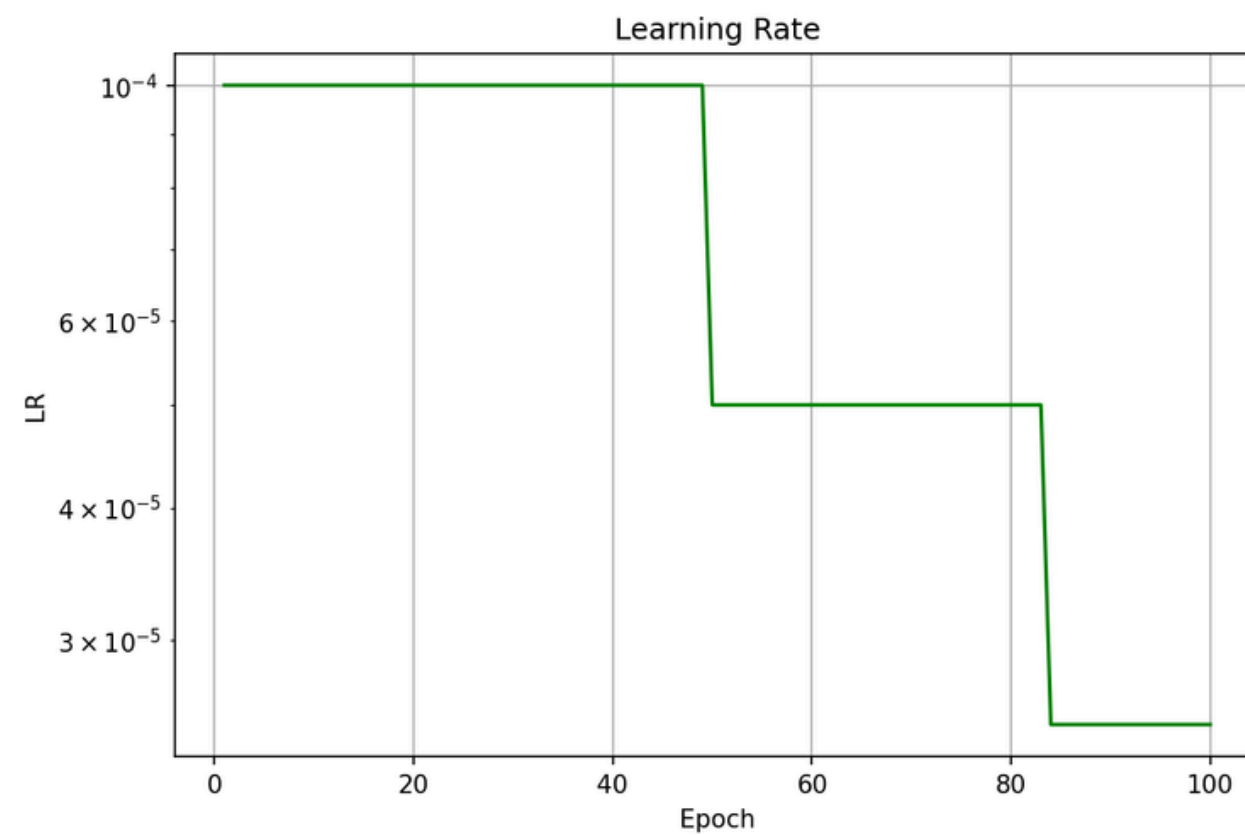
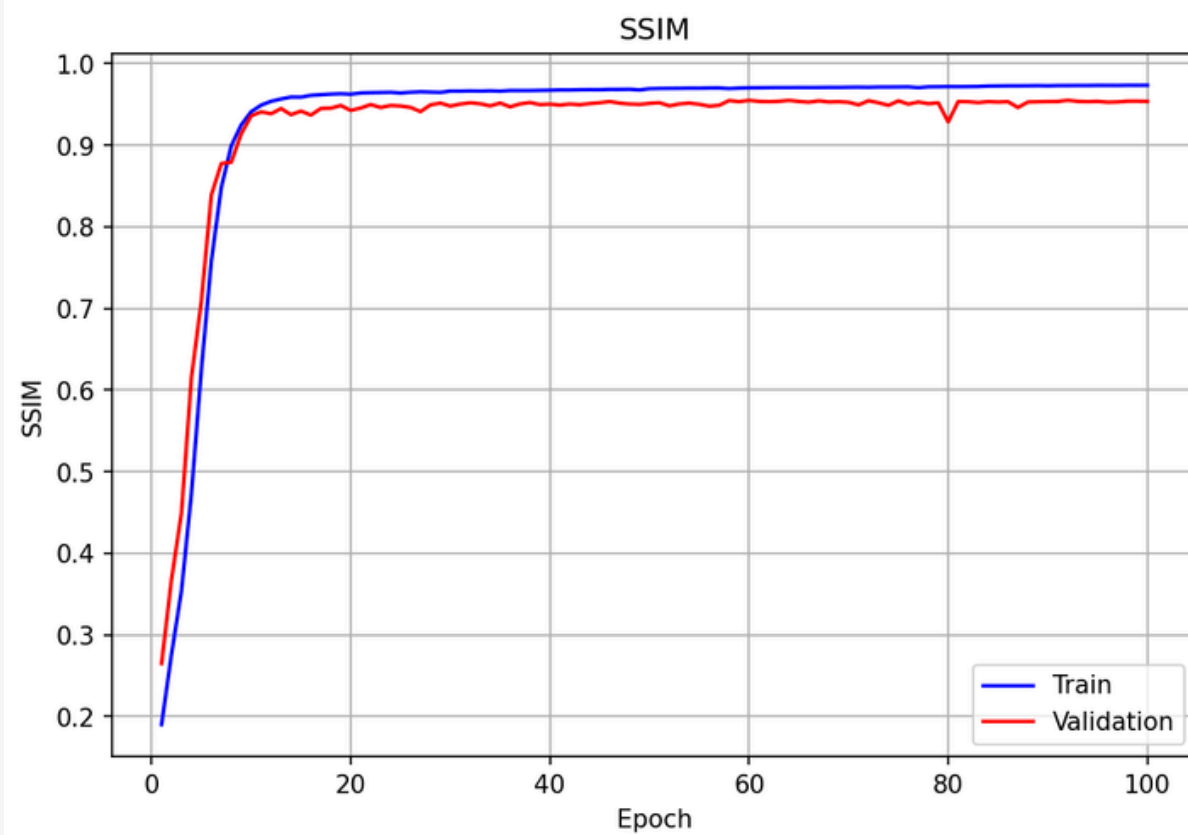
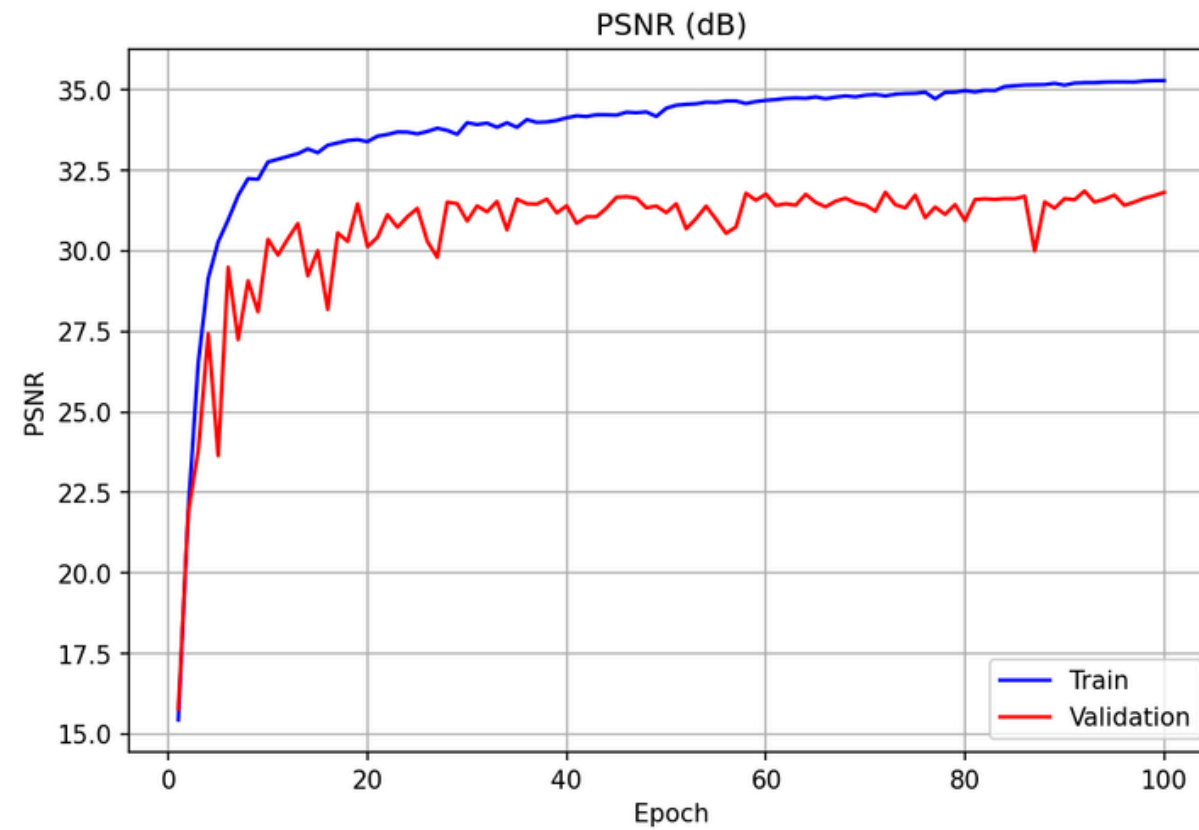
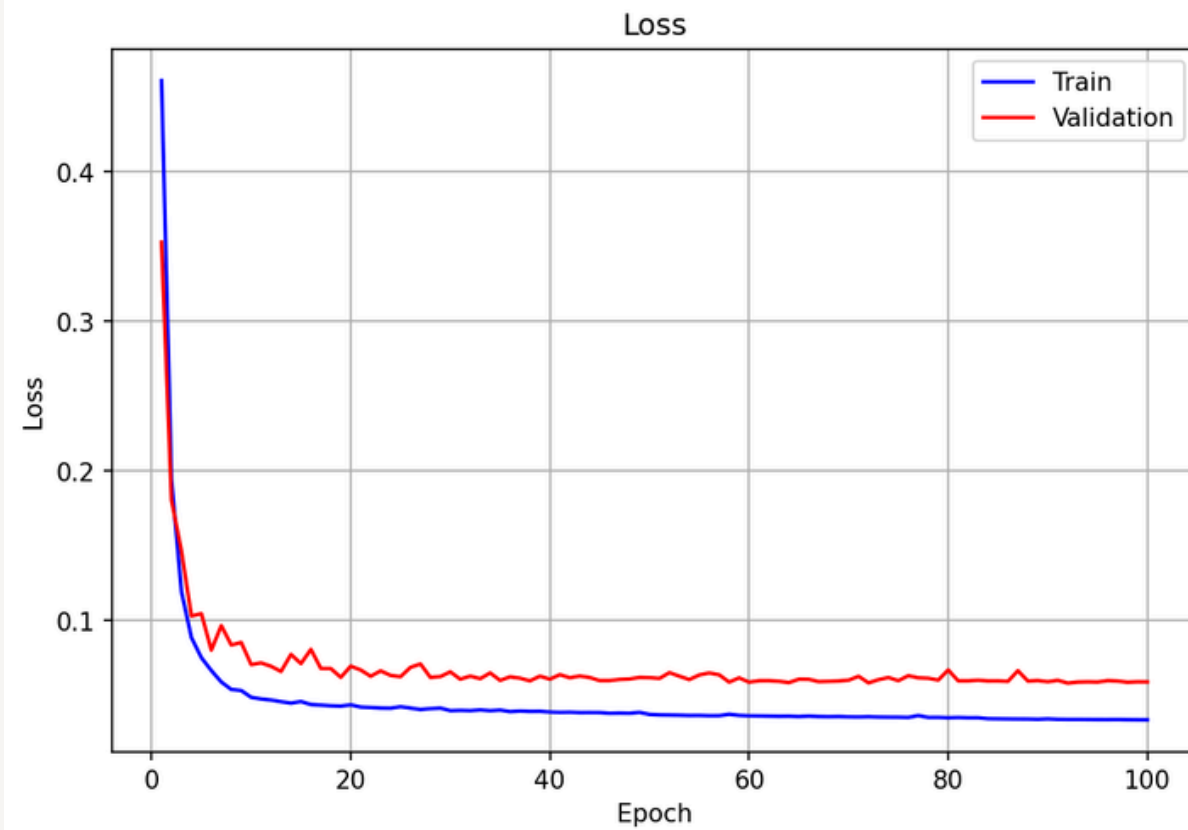
Training Strategy

- Optimized using reconstruction and structural losses
- Stable convergence without adversarial instability
- Trained for fixed epochs on GPU hardware
- Consistent behavior across different patients
- No mode collapse or divergence observed



Mamba Results

Training History - Mamba SSM Dehazing



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MAMBA DEHAZING SUMMARY

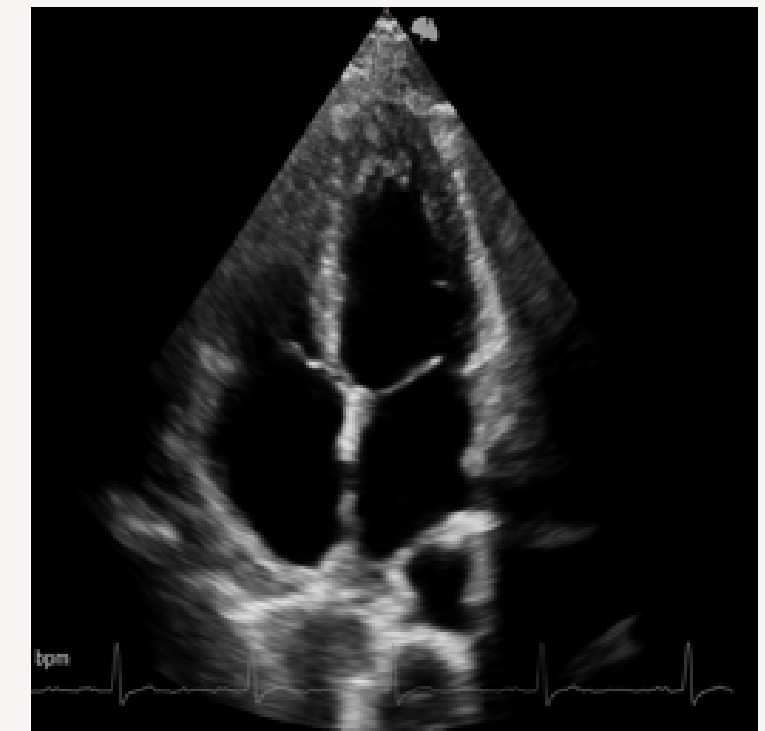
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Best Val PSNR: 31.86 dB (Epoch 92)

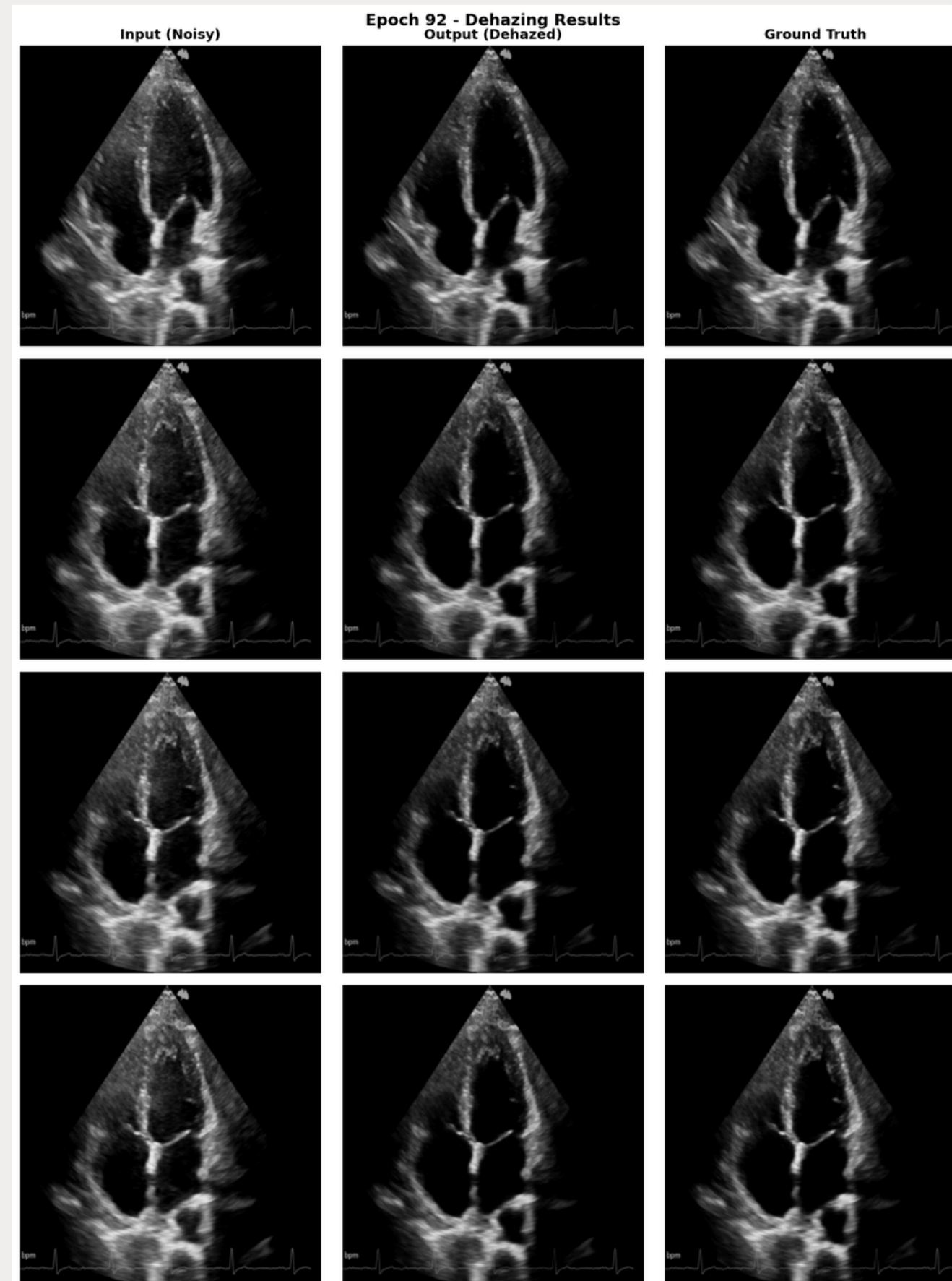
Best Val SSIM: 0.9550

Final MSE: 0.000602

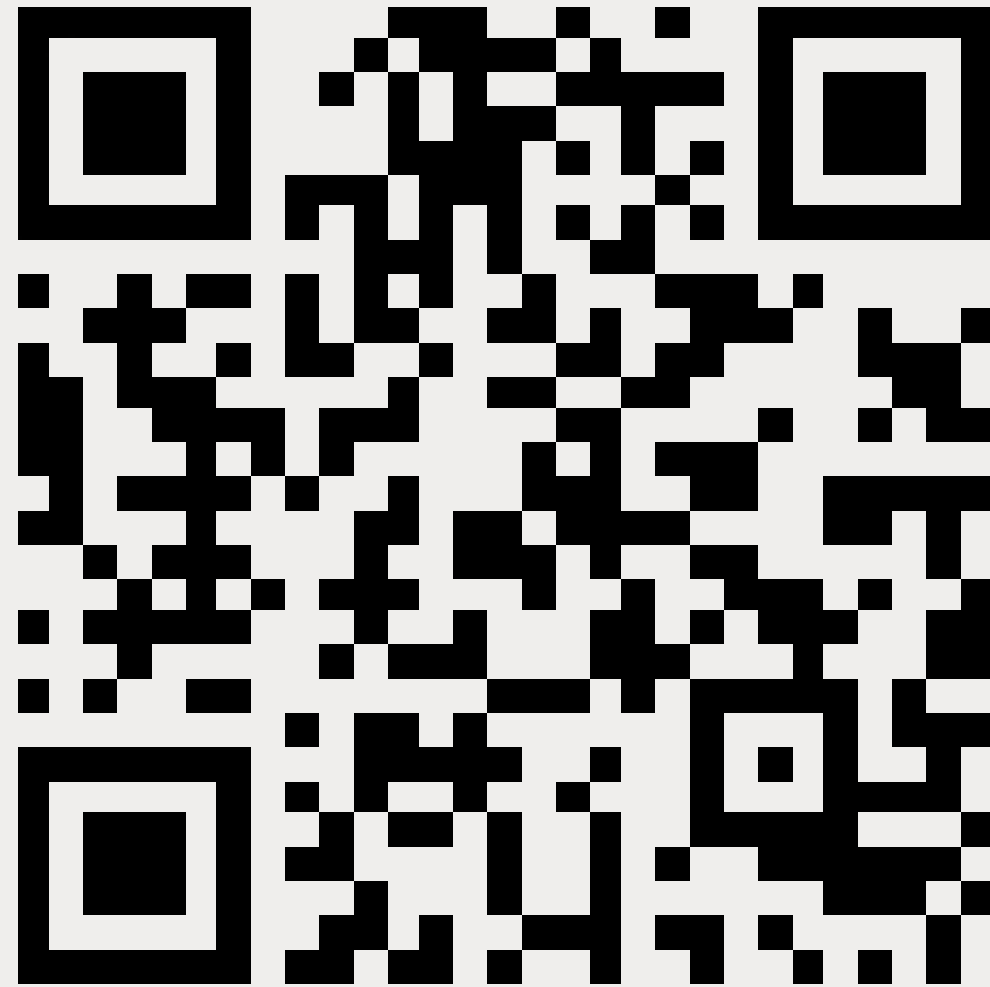
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Output from Mamba



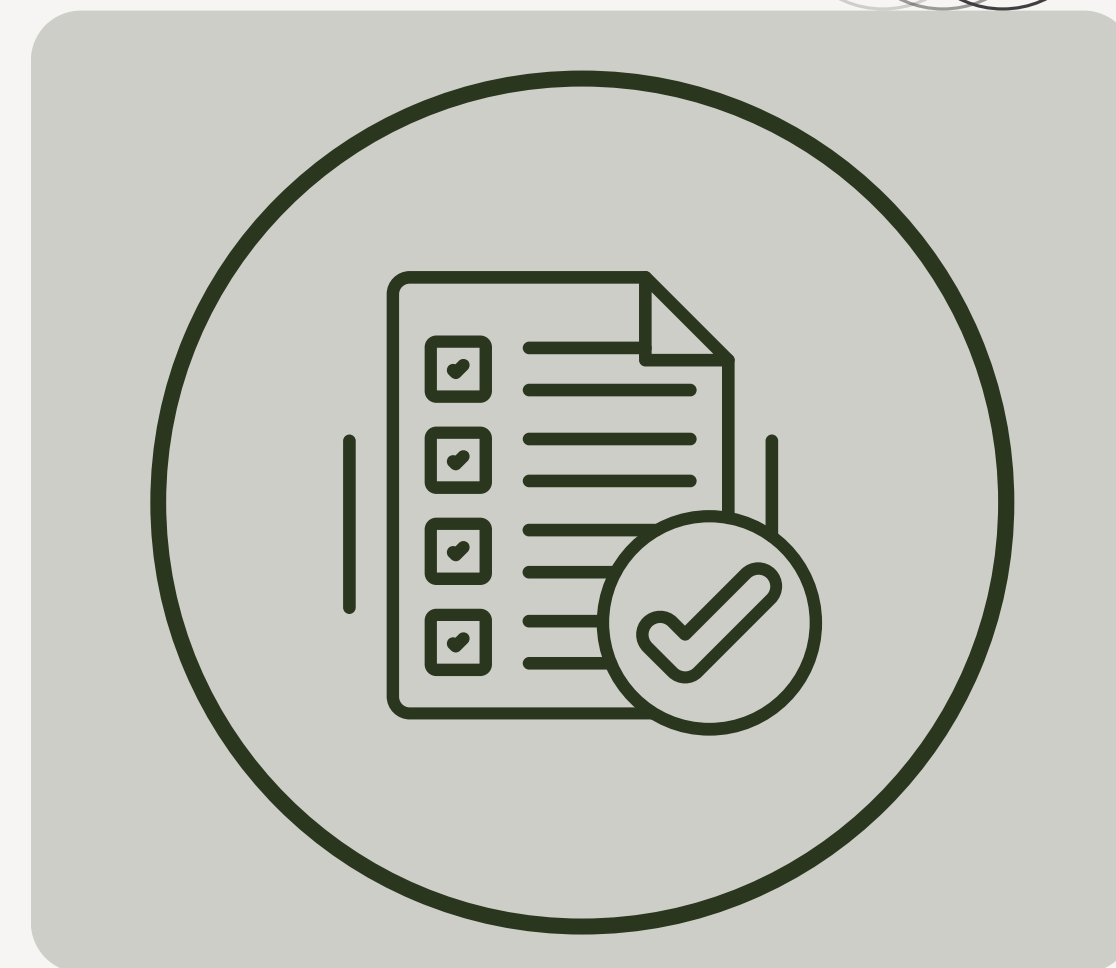
User Interface



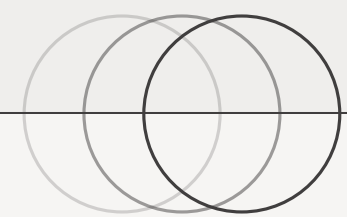
<https://mamba-clarity.pages.dev/>

Conclusion

- Multi-model evaluation confirms strengths and limitations of each approach
- GANs excel in perceptual quality but risk hallucination
- Diffusion models preserve structure but are slow
- Mamba achieves stable, high-fidelity enhancement with clinical safety
- State-space modeling is promising for real-time echocardiographic deployment



Acknowledgements

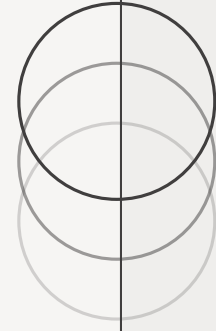


SUPERVISORS

We would like to express our sincere gratitude to **Prof. Dr. M. Monir Uddin**, Professor in the Department of Mathematics & Physics and Founder & Director of Big Matrix Lab at North South University, Dhaka, Bangladesh, for his invaluable guidance, insightful feedback, and continuous support throughout the course of this project.

COLLABORATORS

We are also deeply thankful to our collaborators, **Md. Nishan Khan** and **Kazi Shahriar Sanjid**, for their cordial assistance and contributions at different stages of this work. Their active involvement and support greatly facilitated the completion and success of this project. We also thank all member of **Big-Matrix Lab**



THANK YOU!